



LN8825X Datasheet

A Single-Chip Wireless MCU for Wi-Fi and IOT Applications

Revision 1.9

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1 Overview

1.1 Functional Features

- Wifi Features
 - 2.4GHz 802.11 b/g/n
 - Support STA/AP/STA+AP operation modes
 - Support WPA/WPA2
 - 18dBm output power in 11b mode
 - -94dBm sensitivity
- High Integration
 - Integrated ARM® Cortex - M4F
 - Integrated TCP/IP protocol stack
 - Integrated TR switch, balun, LNA, power amplifier and matching network
 - High precision oscillator
 - AES-128\AES-192\AES-256 hardware encryption
 - 256 bits EFUSE
- Interface
 - Single supply voltage with wide range supported (2.6~3.6V)
 - SPI/SDIO/UART interface
 - SPI/I2C master for sensor
 - I2S, PWM and GPIO
 - Multi-channel ADC
 - XIP from QSPI FLASH with high-speed cache
- Low Power
 - 2.5mA in standby mode (DTIM=1, keep connected with AP)
 - 50uA in deep sleep mode(fast wakeup)
 - 20uA in retention mode
- MCU
 - 40/80/160M clock for M4F
 - 296KB on-chip SRAM
 - Up to 4MB Flash for XIP application code
 - SWD debug interface
- Package
 - LN8825A With 1MB SIP Flash: QFN32 5mm*5mm
 - LN8825B With 2MB SIP Flash: QFN32 5mm*5mm

1.2 Block diagram

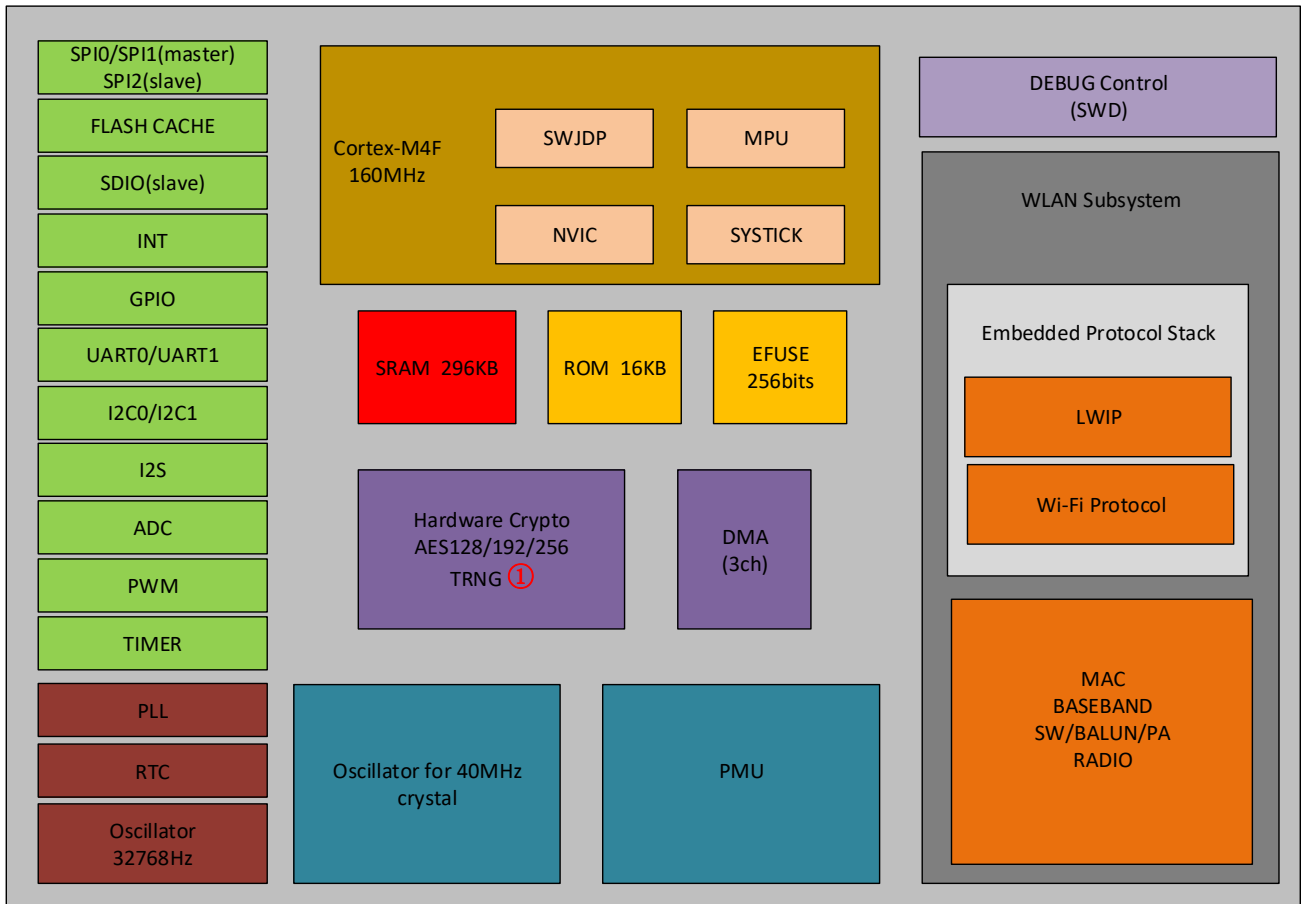


Figure 1-1 LN8825X block diagram

Notes:

- ① TRNG stands for True Random Number Generator

1.3 Memory Map

1.3.1 Memory Map of LN8825A

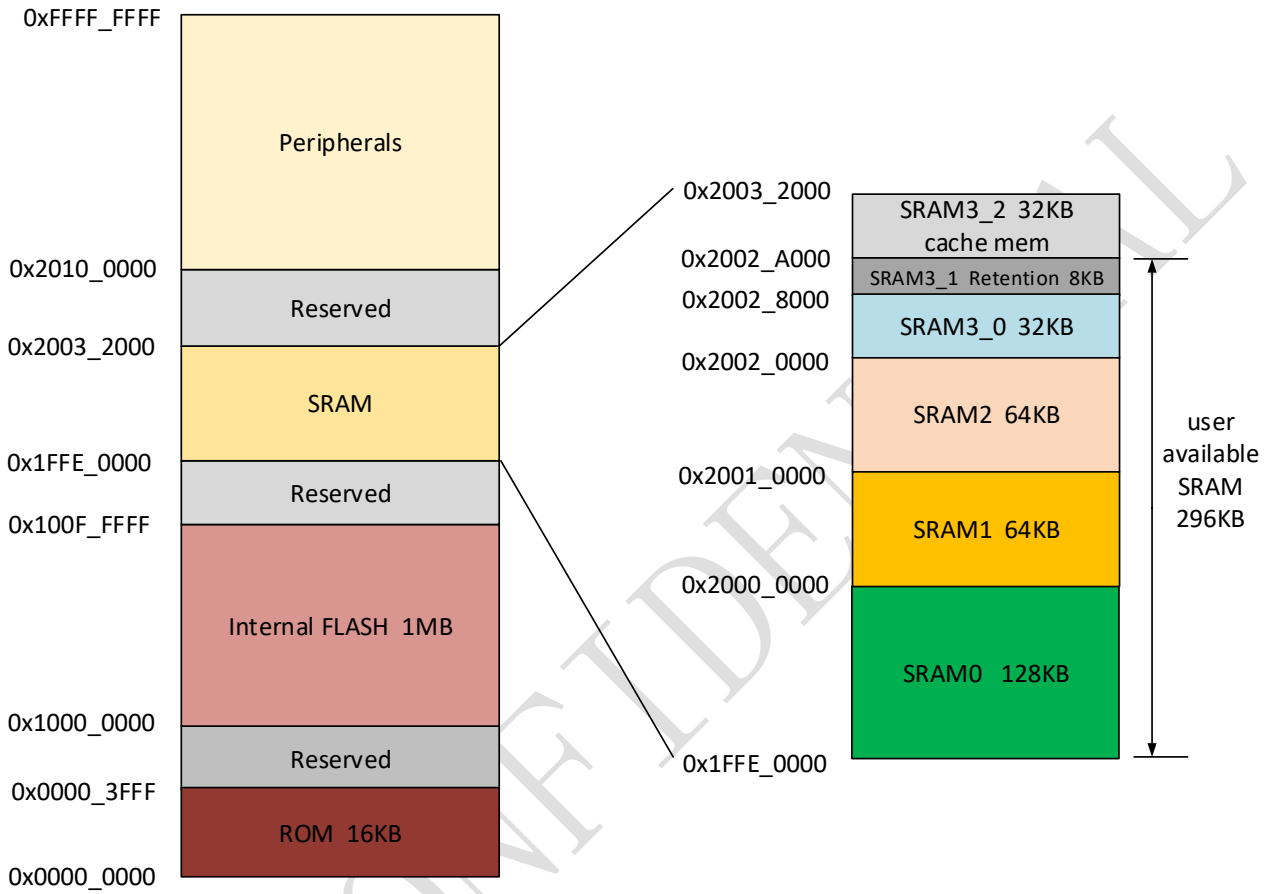


Figure 1-2 LN8825A Memory Map

1.3.2 Memory Map of LN8825B

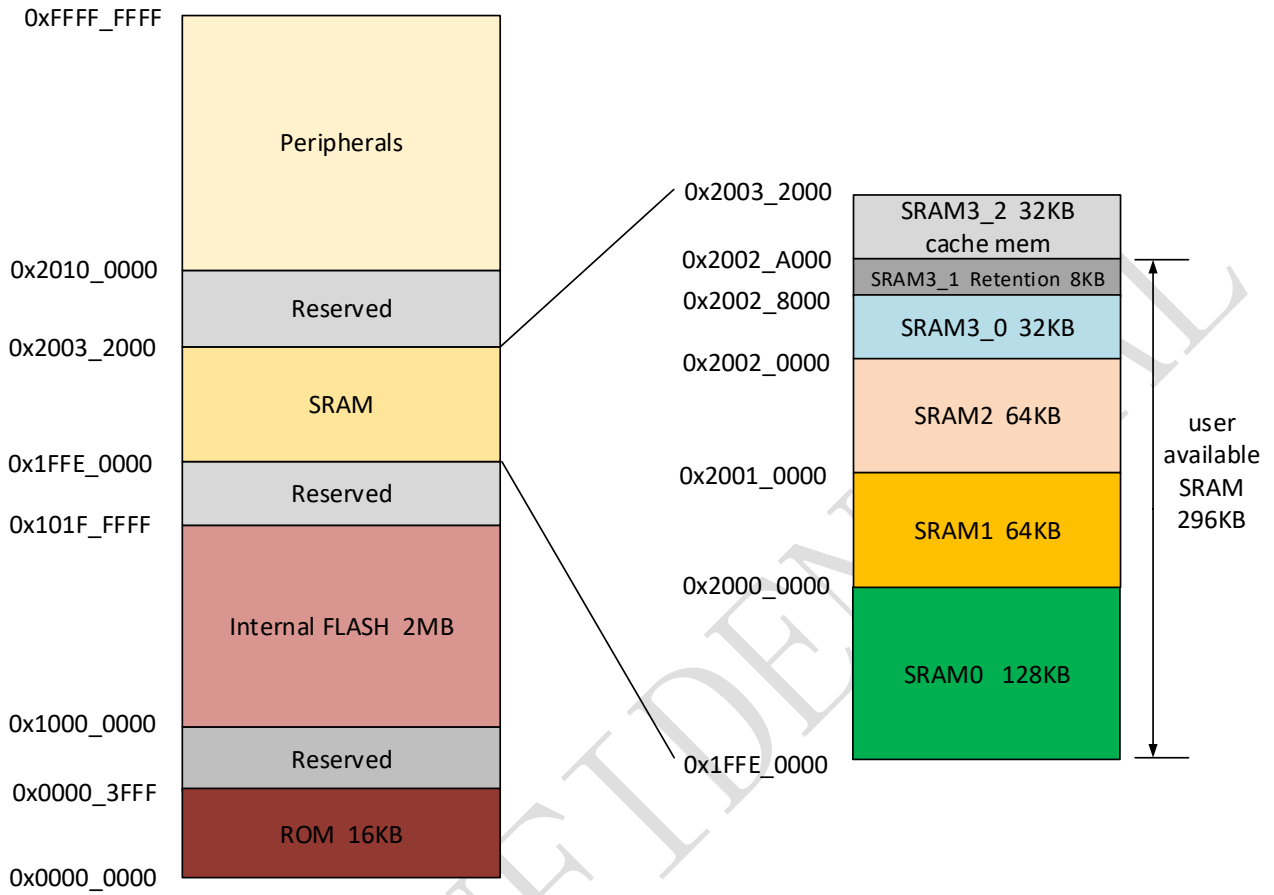


Figure 1-2 LN8825B Memory Map

2 Wi-Fi Subsystem

2.1 Supported Frequencies

Table 2-1 Supported Frequencies

Parameter	Condition	Min	Typ	Max	Unit
Receive frequency range 2.4GHz		2412		2484	MHz

2.2 Receiver Characteristics

Table 2-2 Receiver Characteristics

Parameter	Condition	Min.	Typ.	Max.	Unit
Sensitivity					
11b, 1M	FER<8%, 1024 bytes		-94		dBm
11b, 11M	FER<8%, 1024 bytes		-88		dBm
11g, 6M	FER<10%, 1024bytes		-90		dBm
11g, 54M	FER<10%, 1024 bytes		-74		dBm
11n, MCS0	FER<10%, 1024 bytes		-90		dBm
11n, MCS7	FER<10%, 1024 bytes		-71		dBm
Maximum input level					
11b	FER<8%, 1024 bytes		4		dBm
11g	FER<10%, 1024 bytes		-6		dBm
11n	FER<10%, 1024 bytes		-6		dBm
Operating power consumption					
11b			63		mA
11g			68		mA
11n			68		mA

2.3 Transmitter Characteristics

Table 2-3 Transmitter Characteristics

Parameter	Condition	Min.	Typ.	Max.	Unit
Output power					
11b, 1M DSSS	Maximum RMS output power measured		18	19	dBm
11g, 54M OFDM	Maximum RMS output power measured		14	15	dBm
11n, MCS7	Maximum RMS output power measured		13	14	dBm
Power consumption					
11b	Continuous transmitting		260		mA
11g	Continuous transmitting		220		mA
11n	Continuous transmitting		220		mA

3 Power Management

3.1 Power Mode

There are 5 power modes to manage lower power operation of the system.

- **Active:** Normal mode
SW can gate clock of some Peripherals.
- **LightSleep:** CPU executes WFI instruction without ‘SLEEPDEEP’, and gates clock by itself
Other modules are still in free running.
CPU exit this mode from wakeup by interrupt.
- **DeepSleep:** CPU executes WFI instruction with ‘SLEEPDEEP’.
All digital parts are all clock gated except AWO[1] domain but CPU power is still on.
Voltage supplied to Digital parts can be reduced to save power.
Clock in AWO is switched to 32KHz
WIC wakes up CPU, then all clock and power recover to normal state.
- **Retention:** CPU executes WFI instruction with ‘SLEEPDEEP’
All digital parts are power-down except AWO domain and selected retention SRAM.
Registers inside CPU can be retained
Clock in AWO will be switched to 32KHz
WIC wakeup CPU, then all clock and power recover to normal state.

Any low power mode can't be switched to another low power mode directly, it must be switched to active mode before going to another one.

Clock and power gating of one domain should be configured before executing WFI.

Power Save Mode Current

Table 3-1 Current in WiFi Power Save Mode

Mode	Condition	Min	Typ	Max	Unit
Power Save Mode	DTIM=1 ^[2]		2.5		mA
	DTIM=3		1		mA

Notes:

[1]AWO: The block whose power is never down is named as Always On

[2]Test condition: keep connected with AP

DTIM=n means STA wakes up to listen to beacon from AP in every n times of DTIM interval

3.2 Power Supplies

Table 3-2 Power Supplies

Power Domain	Description	Min	Typ.	Max	Unit
VDD33	Supply for all chips except PA	2.6	3.3	3.6	V
VDDA33	Supply for PA	2.6	3.3	3.6	V
VIO1	Supply for I/O	2.6	3.3	3.6	V
VIO2	Supply for I/O	2.6	3.3	3.6	V

4 Package Specification

4.1 Pin Layout

LN8825X is assembled 5mm x 5mm QFN package of 32-pin with 0.5mm pitch.

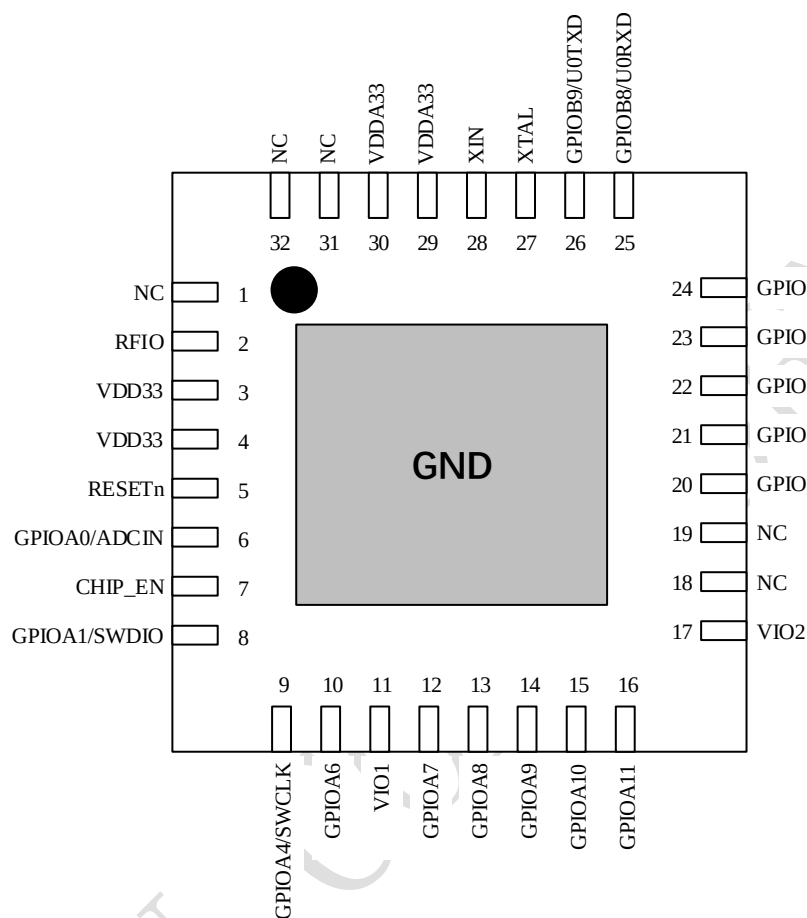


Figure 4-1 LN8825X Pin Layout

4.2 Pin Description

LN8825X totally has 16 General Purpose Inputand Outputs (I/O).

- Two group IOs GPIOA and GPIOB, which are separately configurable
- Configurable hardware and software control for each signal
- Configurable interrupt mode for Port A
- Debounce logic for interrupts
- Generate single hi-active interrupts

4.2.1 Programmable I/Os

Table 4-1 Programmable I/Os

Pin No.	Pin Name	Normal Mode (after reset)	Logic Level	PU / PD / HiZ (Pull 110KΩ) (after reset)	IN/OUT Put (after reset)
6	GPIOA0	GPIOA0	VIO1	PU	IN
8	GPIOA1	SWDIO ^①	VIO1	PU	IN
9	GPIOA4	SWCLK ^②	VIO1	DOWN	IN
10	GPIOA6	GPIOA6	VIO1	DOWN	IN
12	GPIOA7	GPIOA7	VIO1	DOWN	IN
13	GPIOA8	GPIOA8	VIO1	DOWN	IN
14	GPIOA9	GPIOA9	VIO1	PU	IN
15	GPIOA10	GPIOA10	VIO1	PU	IN
16	GPIOA11	GPIOA11	VIO1	DOWN	IN
20	GPIOB3	GPIOB3	VIO2	PD	IN
21	GPIOB4	GPIOB4	VIO2	PU	IN
22	GPIOB5	GPIOB5	VIO2	PU	IN
23	GPIOB6	GPIOB6	VIO2	PU	IN
24	GPIOB7	GPIOB7	VIO2	PU	IN
25	GPIOB8	UART0_RX ^③	VIO2	PU	IN
26	GPIOB9	UART0_TX ^④	VIO2	PU	IN

Notes:

- (1) ^①^② is gpio mode after system reset and can be configured as SWD by bootrom program
 (2) ^③^④ is gpio mode after system reset and can be configured as UART0 by flash program.

4.2.2 Power,Clocks and Reset I/Os

Table 4-2 Power,Clocks and Reset I/Os

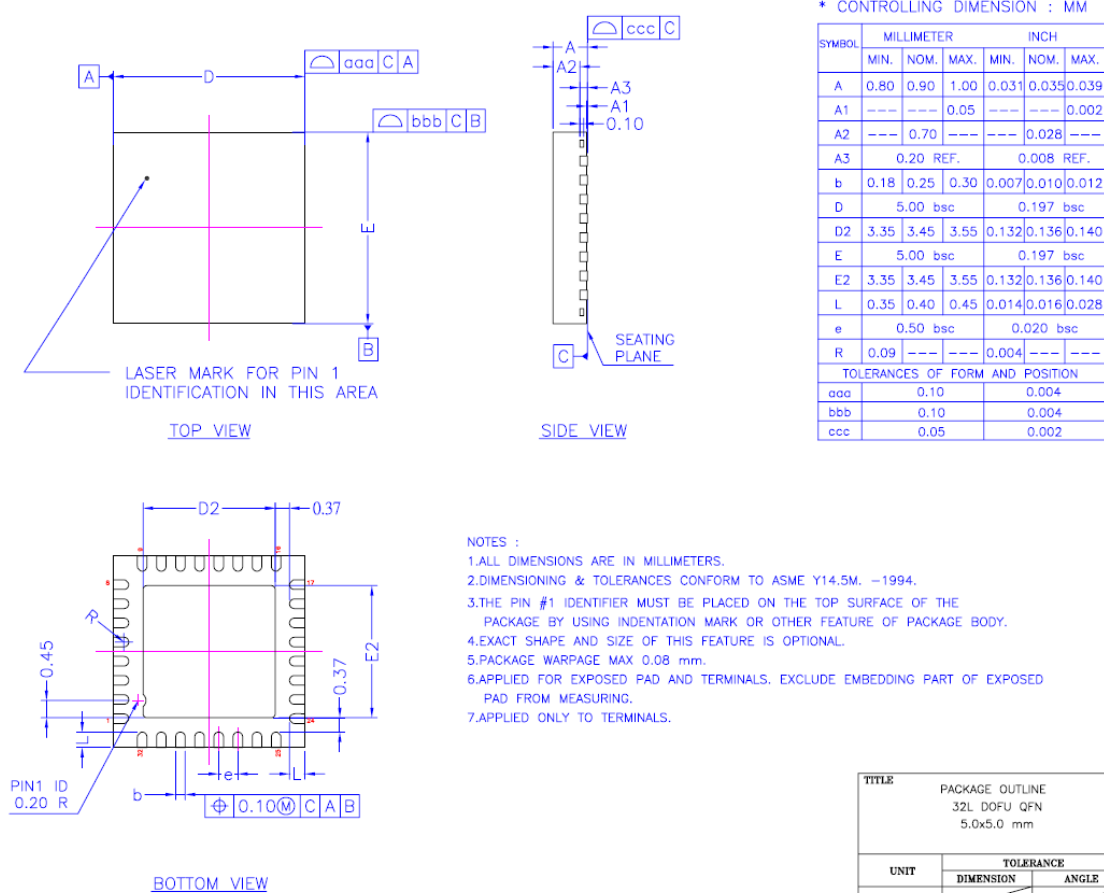
Pin No.	Pin Name	Pin description
1	NC	not connected
2	RFIO	Wi-Fi RF transmitter/receiver
3	VDD33	3.3v power supply
4	VDD33	3.3v power supply
5	RESETn	system reset
7	CHIP_EN	chip enable
11	VIO1 ①	I/O power supply 1
17	VIO2 ②	I/O power supply 2
18	NC	not connected
19	NC	not connected
27	XTAL	Crystal output
28	XIN	Crystal input
29	VDDA33	power supply for PA
30	VDDA33	power supply for PA
31	NC	not connected
32	NC	not connected

Notes:

(1) ① ② VIO1,VIO2 must be supplied externally

(2) ② VIO2 should be the same as the voltage supplied for SPI flash, 3.3V is recommended

4.3 Package Information



5 Peripheral Interface

5.1 SWD

LN8825X supports Serial Wire Debug (SWD) which is used to download flash image and debug. The pins shown in Table 5-1 are assigned to SWD mode by bootrom program after power on.

Table 5-1 SWD Pin Assignment

Pin No.	Pin Name	Function 1
8	GPIOA1	SWD
9	GPIOA4	SWCK

5.2 ADC

LN8825X has a 6-channel ,12-bits sampling Analog-to-Digital Converter (ADC). The main features:

- Up to 6 external channels
- Supply voltage and internal temperature is also measured by the ADC
- Support up to eight result FIFO with selectable FIFO depth
- Conversion complete flag and interrupt
- Selectable asynchronous hardware conversion trigger
- Automatic comparison against programmable value to generate interrupt

Table 5-2 ADC Pin Assignment

Pin No.	Pin Name	Function 2(ADC)
6	GPIOA0	ADC CH0
8	GPIOA1	ADC CH1
9	GPIOA4	ADC CH2
20	GPIOB3	ADC CH3
21	GPIOB4	ADC CH4
22	GPIOB5	ADC CH5

5.3 SDIO

LN8825X features one slave sdio interface, which supports the standard SDIO version 2.0 and provides high-speed data I/O with low power consumption for mobile electronic devices. The main features:

- DMA supported
- The clock is up to 40MHz
- Supported 1-bit, 4-bit data mode

Table 5-3 SDIO Pin Assignment

Pin No.	Pin Name	Function 5(SDIO)
10	GPIOA6	SDIO_IO2
12	GPIOA7	SDIO_IO3
13	GPIOA8	SDIO_CMD
14	GPIOA9	SDIO_CLK
15	GPIOA10	SDIO_IO0
16	GPIOA11	SDIO_IO1

5.4 I2S

LN8825X features one slave I2S interface, which is generally used to handle the transfer of audio data. The pin definition of I2S is shown in Table 5-4.

- I2S transmitter and/or receiver based on the Philips I2S serial protocol
- Slave mode of operation
- 1 channels for both transmitter and receiver
- APB data bus widths of 8, 16, and 32 bits
- Audio data resolutions of 12, 16, 20, 24, and 32 bits
- External sclk gating and enable signals
- Support for programmable DMA registers

Table 5-4 I2S Pin Assignment

Pin No.	Pin Name	Function 6(I2S)
10	GPIOA6	I2S_SDI0
13	GPIOA8	I2S_WS
14	GPIOA9	I2S_CLK
15	GPIOA10	I2S_SDO0

5.5 Interrupt

There are 9 interrupt sources from external pins and 5 interrupt sources that can wake up the system from sleep. These interrupts support both edge and level mode. The assignment for interrupt pins is listed in Table 5-5.

Table 5-5 Interrupt Pin Assignment

Pin No.	Pin Name	Function 9(EXT INT)	Wakeup INT
6	GPIOA0	EXT INT0	WAKEUP INT[0]
8	GPIOA1	EXT INT1	WAKEUP INT[1]
9	GPIOA4	EXT INT4	
10	GPIOA6	EXT INT6	WAKEUP INT[5]
12	GPIOA7	EXT INT7	WAKEUP INT[6]
13	GPIOA8	EXT INT8	
14	GPIOA9	EXT INT9	
15	GPIOA10	EXT INT10	
16	GPIOA11	EXT INT11	
26	GPIOB9		WAKEUP INT[7]

5.6 Boot Mode

LN8825X has two boot modes that depend on the logic value set of GPIOA8, GPIOA9 and GPIOA10 during power-on stage. GPIOA8 is pulled down while GPIOA9 and GPIOA10 are pulled up internally. The pin assignment is listed in Table 5-6 and boot mode configuration is shown in Table 5-7.

Table 5-6 Boot Mode Pin Assignment

Pin No.	Pin Name	Function 10(Boot Mode)
13	GPIOA8	BOOT_MODE[0]
14	GPIOA9	BOOT_MODE[1]
15	GPIOA10	BOOT_MODE[2]

Table 5-7 Configuration of Boot Mode

MODE	GPIOA8	GPIOA9	GPIOA10	Description
FLASH Boot	X	X	1	DEFAULT: GPIO8 pull-down GPIO9 and GPIO10 pull-up internally.
UART Boot	0	1	0	CPU will wait data from UART to program flash.

5.7 Alternate Function I/O

The pins that are named as alternate function I/O can be configured to any function mode at one time. There are 16 alternate functional I/Os referred to Table 5-8 and 38 function modes shown in Table 5-9.

Table 5-8 Alternate function I/Os

Pin No.	Pin Name	Function 7(Alternate Function IO)
6	GPIOA0	AltFunIO[0]
8	GPIOA1	AltFunIO[1]
9	GPIOA4	AltFunIO[4]
10	GPIOA6	AltFunIO[6]
12	GPIOA7	AltFunIO[7]
13	GPIOA8	AltFunIO[8]
14	GPIOA9	AltFunIO[9]
15	GPIOA10	AltFunIO[10]
16	GPIOA11	AltFunIO[11]
20	GPIOB3	AltFunIO[13]
21	GPIOB4	AltFunIO[14]
22	GPIOB5	AltFunIO[15]
23	GPIOB6	AltFunIO[16]
24	GPIOB7	AltFunIO[17]
25	GPIOB8	AltFunIO[18]
26	GPIOB9	AltFunIO[19]

Table 5-9 Alternate functions

Function Num	Function Mode	Function Num	Function Mode
0	I2C0_SCL	19	PWM5
1	I2C0_SDA	20	PWM6
2	I2C1_SCL	21	PWM7
3	I2C1_SDA	22	PWM8
4	UART0_TX	23	PWM9
5	UART0_RX	24	PWM10
6	UART0_RTS	25	PWM11
7	UART0_CTS	26	SPI1_SCK ^⑤
8	UART1_TX	27	SPI1_CS
9	UART1_RX	28	SPI1_MOSI
10	TIMER_PWM0	29	SPI1_MISO
11	TIMER_PWM1	30	SPI2_CLK ^⑥
12	TIMER_PWM2	31	SPI2_CS
13	TIMER_PWM3	32	SPI2_MOSI
14	PWM0	33	SPI2_MISO
15	PWM1	34	SPI0_CLK ^⑦
16	PWM2	35	SPI0_CS
17	PWM3	36	SPI0_MOSI
18	PWM4	37	SPI0_MISO

Note:

(1). ^{⑤⑦}SPI0 and SPI1 only support master mode;

(2). ^⑥SPI2 only supports slave mode

5.8 I2C

LN8825X has two I2C interfaces which both support master and slave mode. Any two pins listed in [Table 5-8](#) can be configured as I2C. Their features:

- Standard, Fast and high-speed mode are supported
- Master or slave I2C operation
- 7 or 10-bit addressing, 7 or 10-bit combined format transfers
- Bulk transmit mode, DMA request signals
- Programmable slave address, default slave address of 0x055

- Interrupt or polled-mode operation
- Programmable SDA hold time

5.9 UART

There are 2 UARTs in LN8825X, UART0 and UART1. UART0 supports hardware flow control, which is generally used to output log. While UART1 is suggested to be used for transferring data, such as AT command. Any pin listed in [Table 5-8](#) can be configured as UART. Their common features:

- 16 bytes Transmit and receive FIFOs.
- DMA supported
- Shadow registers to reduce software overhead
- Transmitter Holding Register Empty (THRE) interrupt mode
- Functionality based on the 16550 industry standard:
- Programmable character properties, such as number of data bits per character (5-8), optional
- parity bit (with odd or even select) and number of stop bits (1, 1.5 or 2)
- Line break generation and detection
- Prioritized interrupt identification
- Programmable serial data baud rate (up to 2000000 B/S)

5.10 SPI

There are three independent SPI interfaces: SPI0 and SPI1 are masters, and SPI2 is slave. Any pin listed in [Table 5-8](#) can be configured as dedicated function for SPI. There are no FIFO or no DMA supported in SPI1.

Features:

- 4-wire serial interface: SCK CSn SDI SDO
- 16 words Tx and Rx fifo (SPI0/SPI2)
- DMA interface supported (SPI0/SPI2)

- Dynamic control of the serial bit rate of the data transfer. The maximum frequency of the bit-rate clock (sclk_out) is half of APBCLK

5.11 PWM

There are 4 PWMs inside Timer which has 24-bit width counter. Another 12 PWMs with 16-bit width counter are integrated. The clock source is from APB clock. Duration of low level and high level need to be set, and it determines the duty cycle of PWM waveform. Pins listed in [Table 5-8](#) can be configured as output of PWM.

6 Electrical Characteristics

6.1 Absolute Maximum Ratings

Table 6-1 Absolute Maximum Ratings

Symbol	Min.	Max.	Unit
Supply Voltage	0	3.6	V
Logical Voltage	0.3*VIO	VDD33+0.3	V
Storage Temperature	-40	120	°C

6.2 Recommended Operating Conditions

Table 6-2 Recommended Operating Conditions

Symbol	Symbol	Min.	Typ.	Max.	Unit
Power Supply	VDD33	2.6	3.3	3.6	V
VIO	VIO1/VIO2	2.6	3.3	3.6	V
Operating Temperature		-40	25	110	°C

6.3 Input/Output Terminal Characteristics

Table 6-3 Input/Output Terminal Characteristics

Characteristic	Min.	Typ.	Max.	Unit
Input low voltage V _{IL}	-0.3	0	0.6	V
Input high voltage V _{IH}	VIO-0.6	VIO	VIO+0.3	V
Output low voltage V _{OL}	-0.45	0	0.45	V
Output high voltage V _{OH}	VIO-0.5	VIO	VIO+0.5	V

7 Application Notes

7.1 Schematic Guide

- 1) GPIOA8,GPIOA9 and GPIOA10 need to be kept unconnected or pulled according to different boot mode
- 2) CHIP_EN must be pulled or driven to high to enable the chip. It can be used to power down the entire chip by pulling it low
- 3) The VIO must be supplied externally or tied with the VDD33
- 4) Confirm the key components,refers to Table 7-1
- 5) RESETn is recommended to be connected to ground through a 2.2nF capacitor

Table 7-1 Key Components

Item	Part Number	Description	Vendor	Comment
Crystal	TZ0308D	SMD;40M;3.2x2.5mm; -30C ~ +85C;CL=9Pf;+/-10ppm	TST	
Flash	GD25Q16ETIG	16Mb;3.0V;	GD	High-temperature
Flash	XT25F16B	16Mb;3.0V;	XTX	Normal-temperature

7.2 Layout Guide

- 1) 4 Layers is recommended
- 2) Set Ground Plane closed to component plane
- 3) Keepout the crystal signal PINs image on Ground plane
- 4) No slot on the RF return path
- 5) Keep the power trace width above 15mil
- 6) Keep the complex impedance of RF path 50 Ohm

8 Revision History

Table 8-1 Revision History

Version	Changlist	Name	Date
1	LN8825X first version	Shaohe.huang	2019/12/04
1.1	Fix wifi and power parameter mistakes	Shaohe.huang	2019/12/05
1.2	Remove chapter 9 Appendix	Shaohe.huang	2019/12/06
1.3	Chapter5.5 Interrupt pin error correction	Shaohe.huang	2019/12/16
1.4	1.Update wifi receive characteristics parameter; 2.Chapter 1.3 Memory Map error correction	Shaohe.huang	2019/12/25
1.5	Update wifi transmitter characteristics parameter	Shaohe.huang	2020/01/07
1.6	Update Chapter of ADC and Alternate Function I/O	Shaohe.huang	2020/01/15
1.7	Update wifi transmitter and receiver characteristics parameter	Shaohe.huang	2020/03/10
1.8	Update wifi receiver characteristics parameter	Shaohe.huang	2020/03/27
1.9	Discription error correction about PU/PD of GPIOB3 and GPIOB7 refered to Chapter 4.2.1	Shaohe.huang	2020/06/09